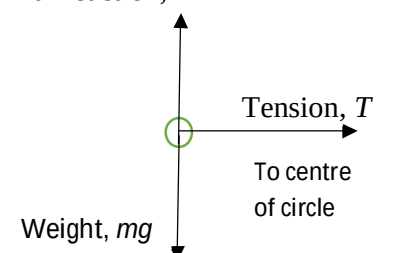
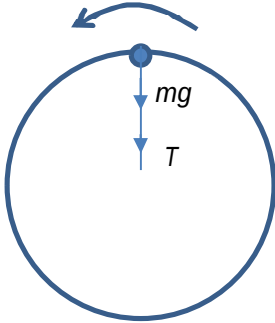


3	(a)	Normal reaction, N  Weight, mg Tension, T To centre of circle Side view of forces acting on the mass.	
	(b)	Tension provides centripetal force for circular motion: $\frac{mv^2}{r} = kx = (40) (0.70 - 0.50) = \mathbf{8.0 \text{ N}}$ $v = \sqrt{\frac{(8.0)(0.70)}{0.050}} = \mathbf{10.6 \text{ m s}^{-1}} = \mathbf{11 \text{ m s}^{-1}}$	[

	(c)	 Consider the object moving in a circular path in a vertical plane. For circular motion of the object, $T + mg = \frac{mv^2}{r}$ where T and m are the tension in the string and the mass of the object respectively. To just become slack, the tension in the string just drops to zero [alternative M1]. Hence $mg = \frac{mv^2}{r}$ $v_{\min} = \sqrt{rg}$ (or $\sqrt{9.81r}$ or $3.13\sqrt{r}$)	

4	(a)	region of space where a magnetic pole (or magnet) moving charged	
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